CNENGR115 Computers and Control

Proportional Action

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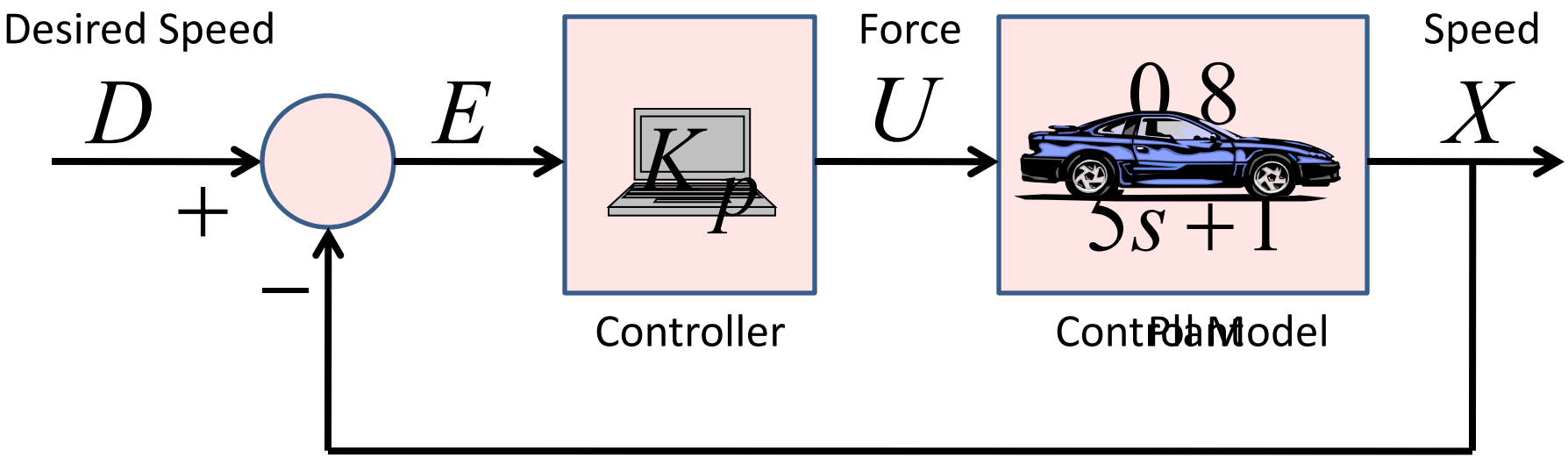
Block Diagram for Closed–Loop Control of Vehicle Speed



“Proportional Control”

Control Lectures

Part 1 – Introduction ✓

Part 2 – Mathematical Models ✓

Part 3 – Control Actions

Proportional

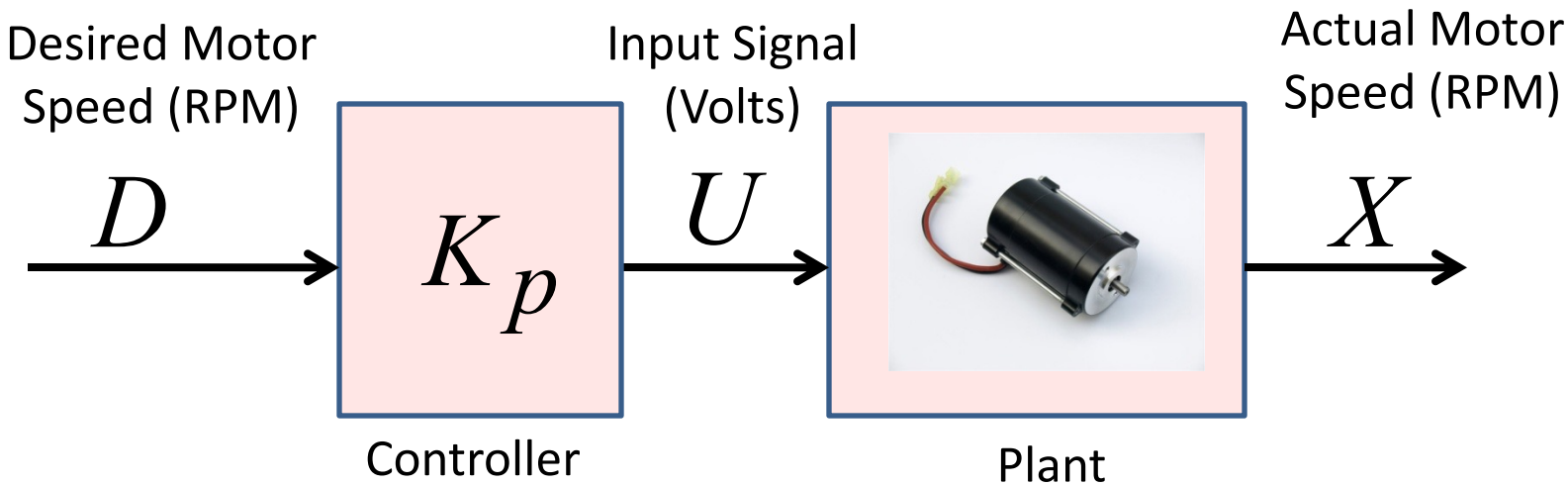
Integral

Proportional + Integral

Control of a DC Motor

- A common actuator is a **DC motor** providing direct rotary motion or, via drums and cables, translational motion
- For control purposes, we will consider the **input voltage** and **shaft speed output**

Open–Loop Control of a DC Motor

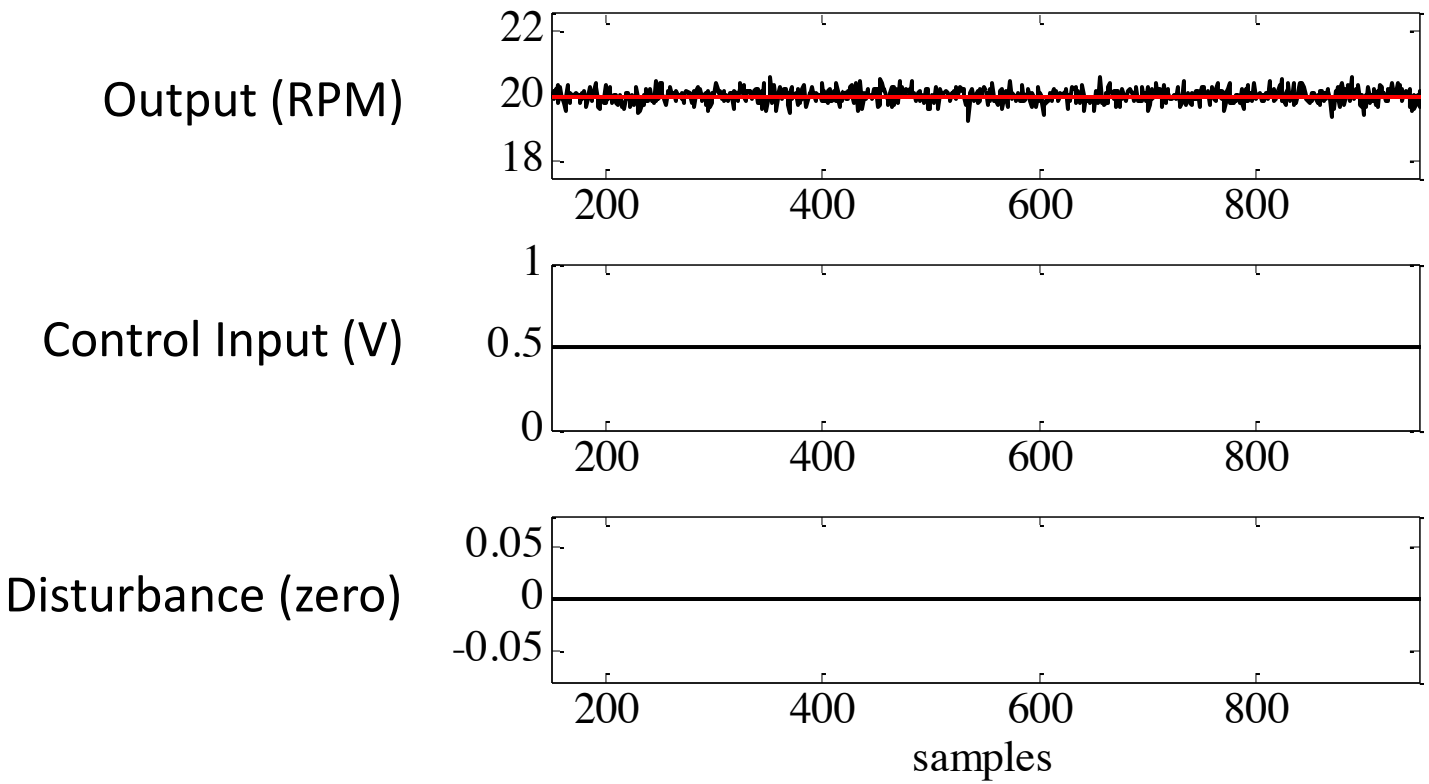


- User specifies a voltage representing the desired motor speed
- Controller is an **electronic amplifier** that determines the input voltage
- For an open-loop controller, the amplifier voltage is:

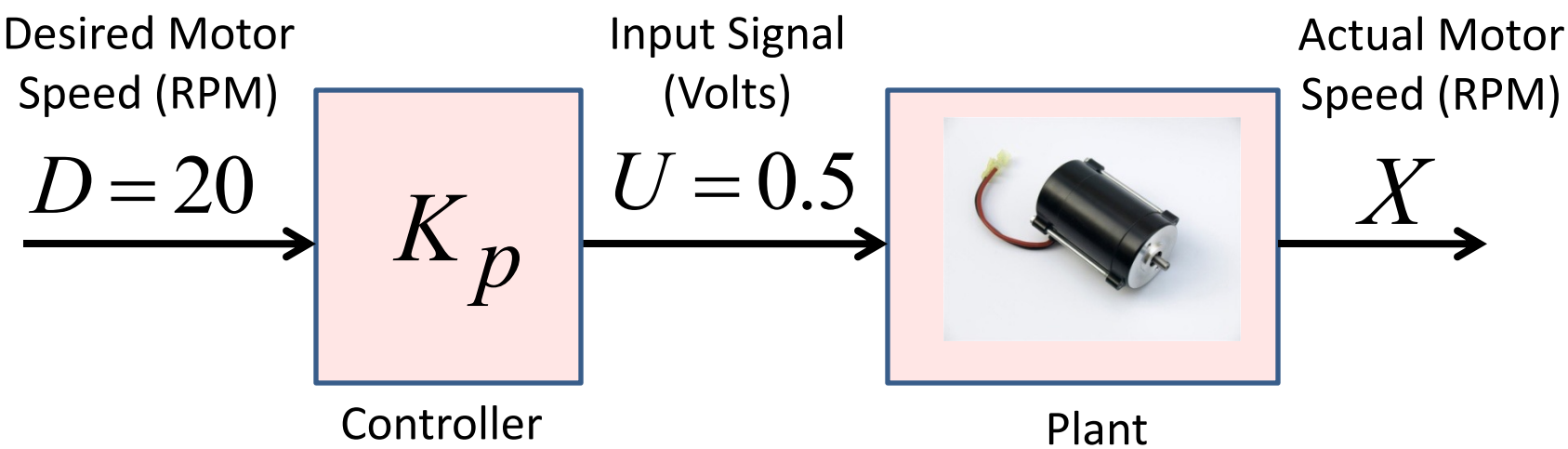
$$U = K_p \times D$$

- However, the actual speed depends on the motor dynamics and any **load disturbances**

Open–Loop Control of a DC Motor



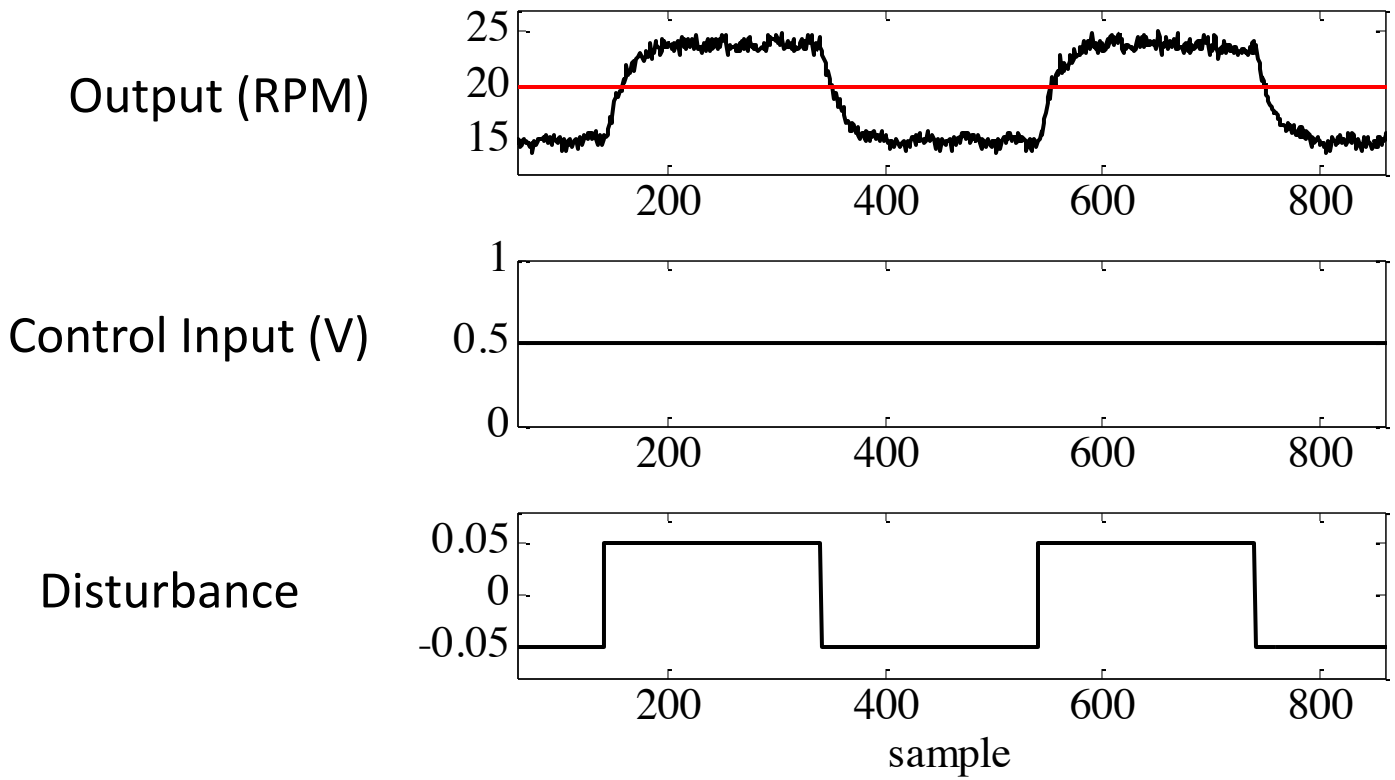
Exercise



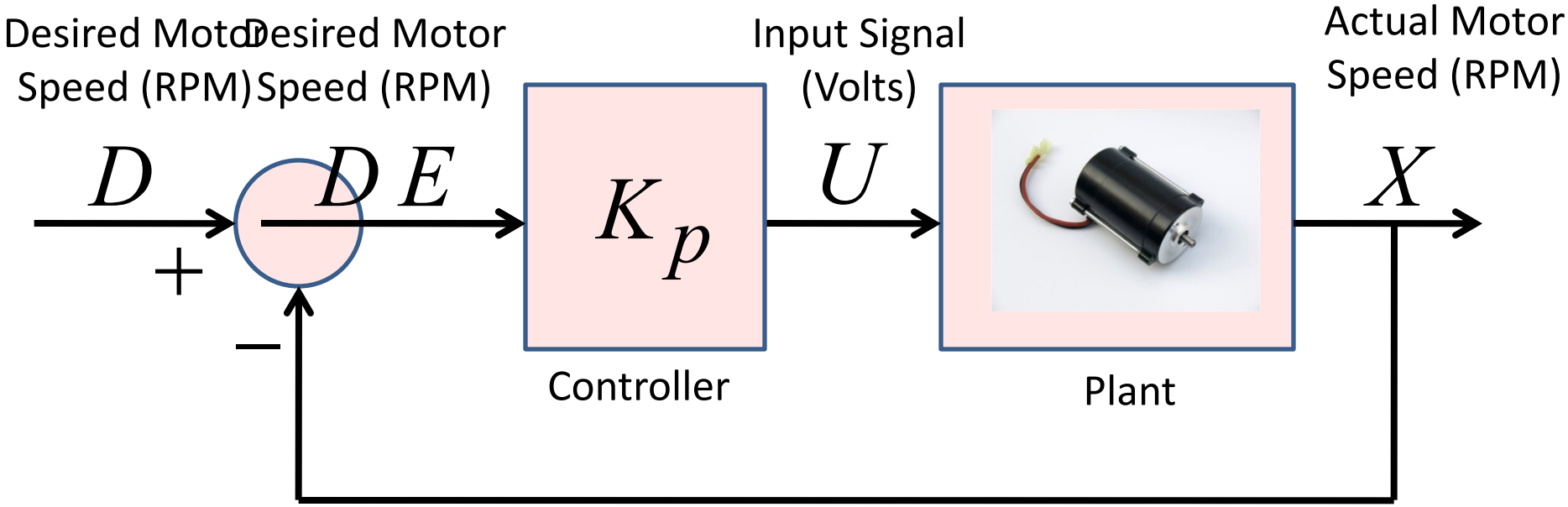
NOT AVAILABLE IN PREVIEW

What value did we use for Kp?

Open–Loop Control of a DC Motor

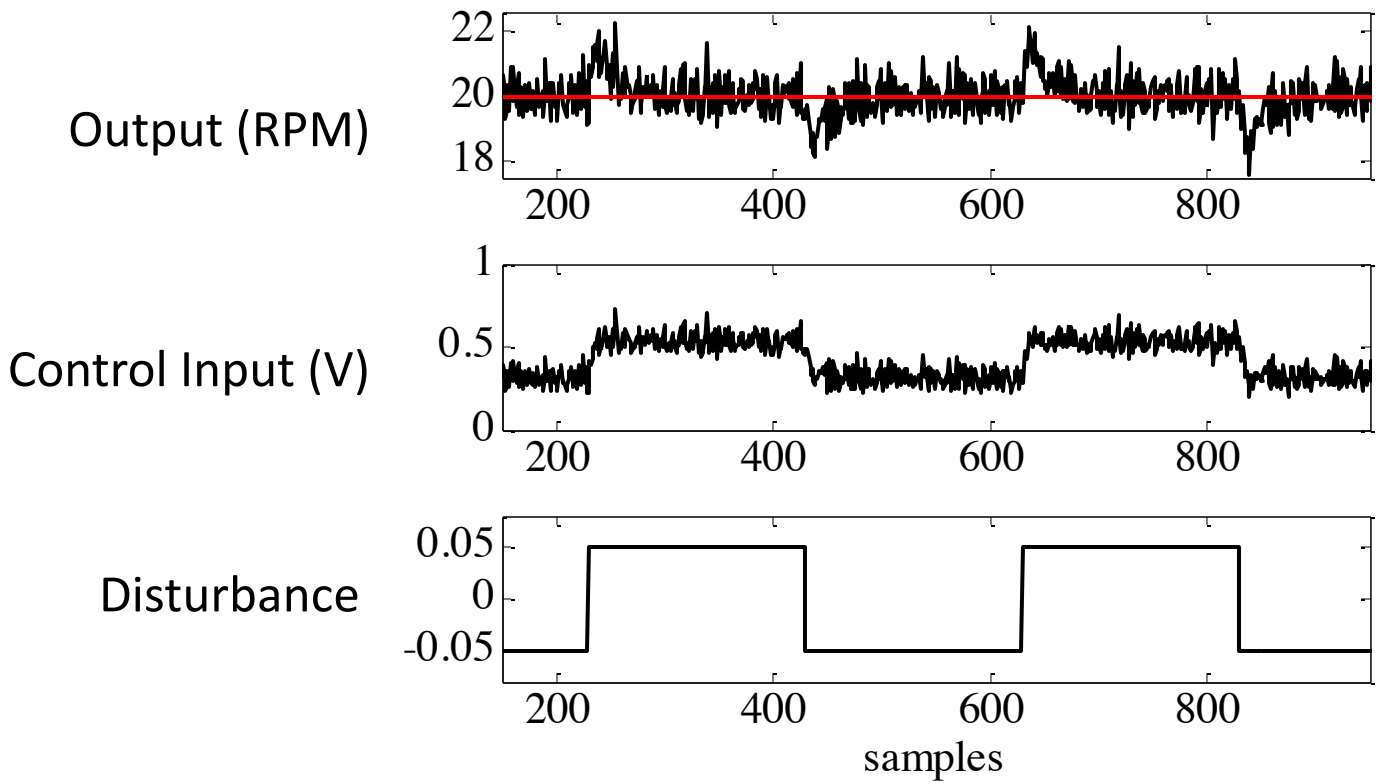


Closed–Loop Control of a DC Motor



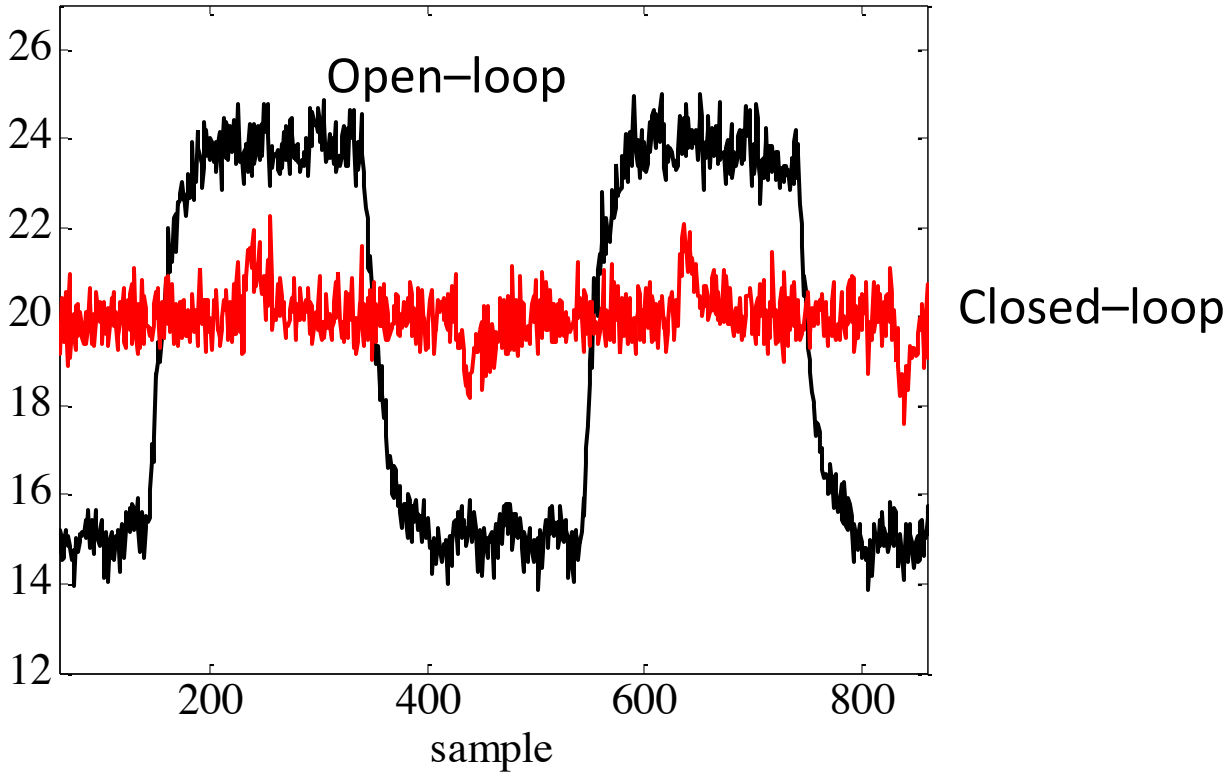
- Output is measured with a **tachometer**, usually a small permanent–magnet that returns a voltage proportional to shaft speed
- Closed–loop controller compares two voltages, i.e. representing the desired and actual speed $E = D - X$
- An electronic amplifier produces a voltage proportional to this error
- **Reduced sensitivity to load disturbances** $U = K_p E = K_p (D - X)$

Closed–Loop Control of a DC Motor



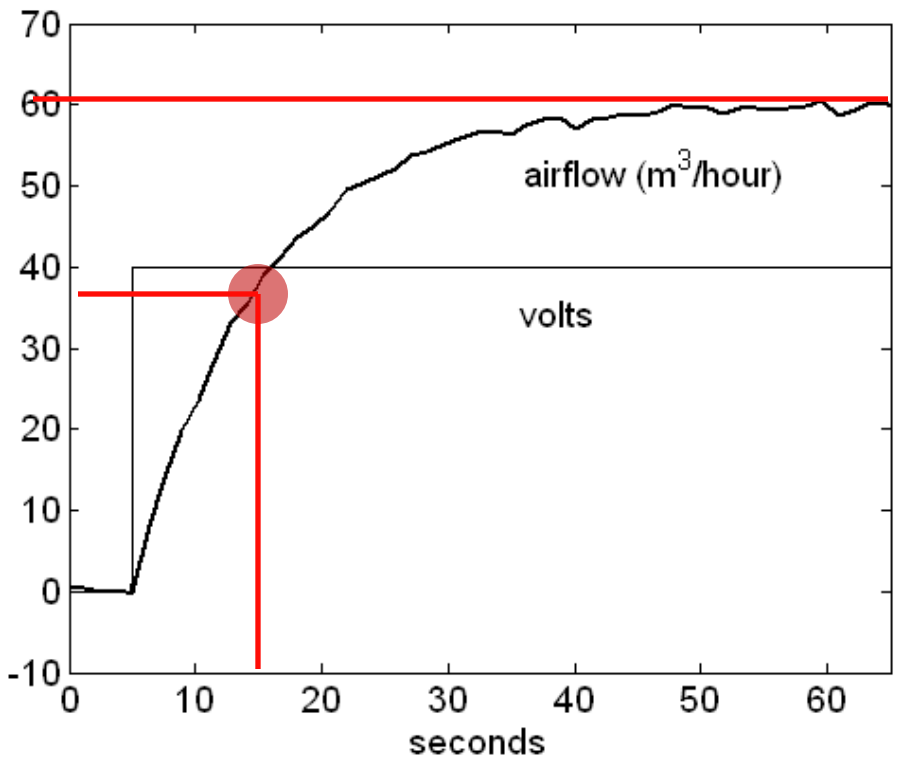
(Actually, this is **proportional + integral** control – see later)

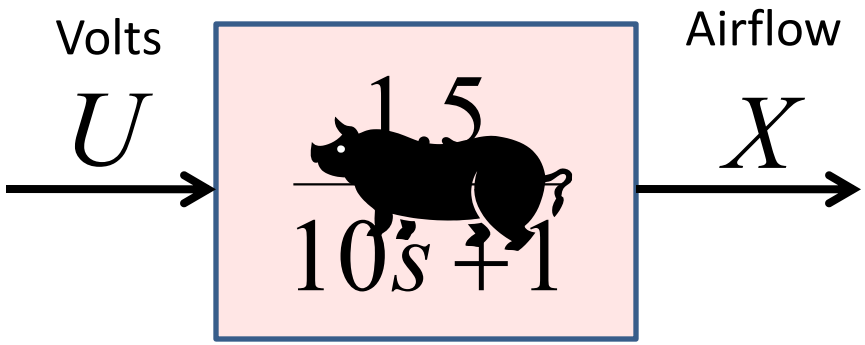
Closed–Loop Control of a DC Motor



Environmental Control

Transfer Function Model for Ventilation Rate





Control Model

The ventilation system includes disturbances, fan dynamics, dimensions of the room, sensors etc.

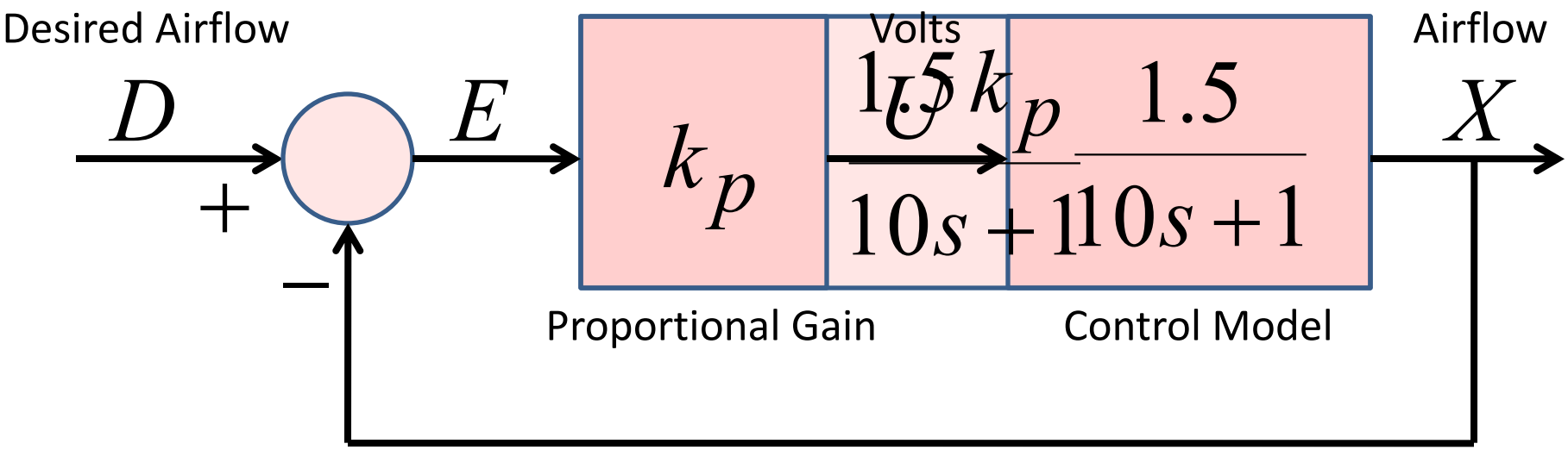
Proportional Control of Ventilation Rate

- The **proportional control law** is given by

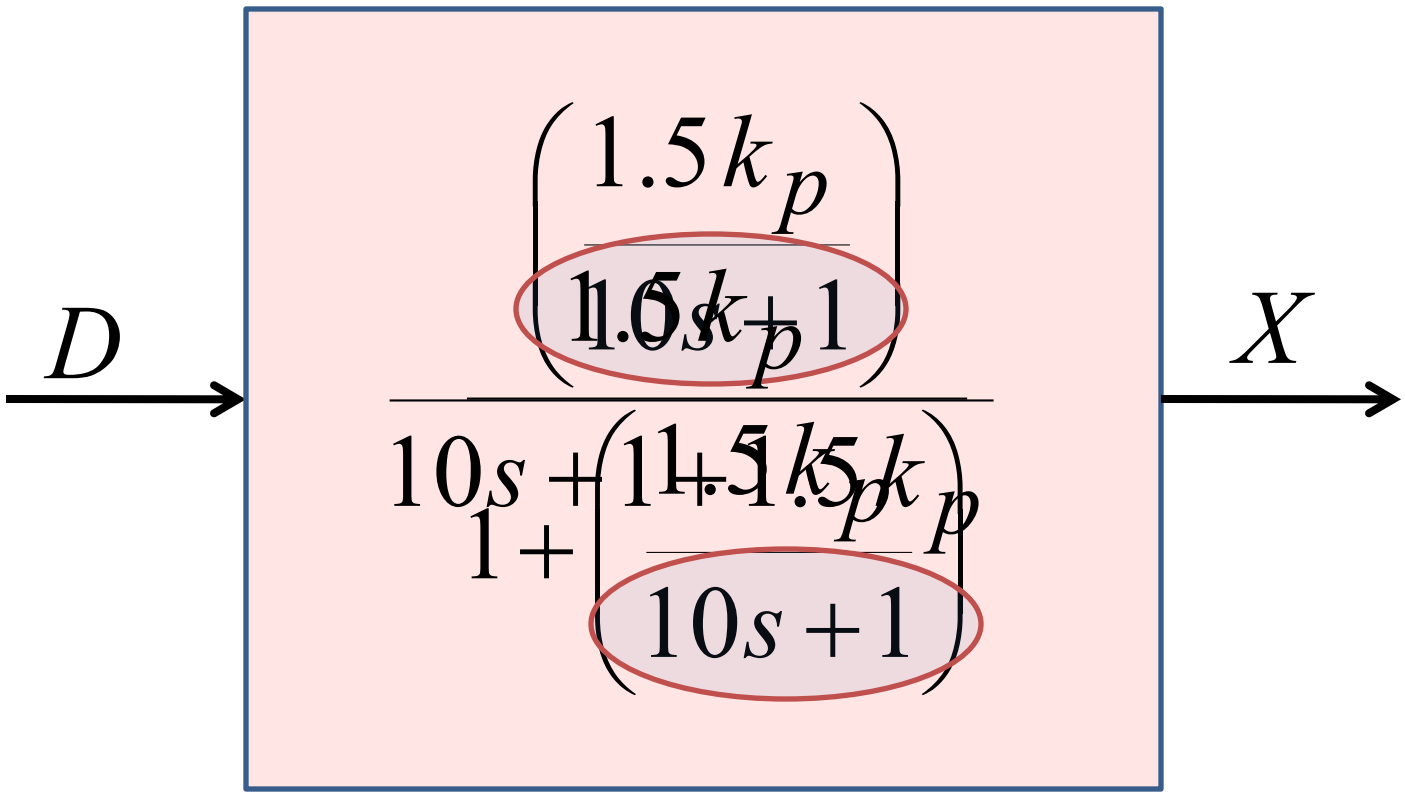
$$U = k_p \left(D - X \right)$$

- At each time instant, the control input is the error signal (the difference between desired and actual ventilation rate) multiplied by a simple gain

Proportional Control of Ventilation Rate



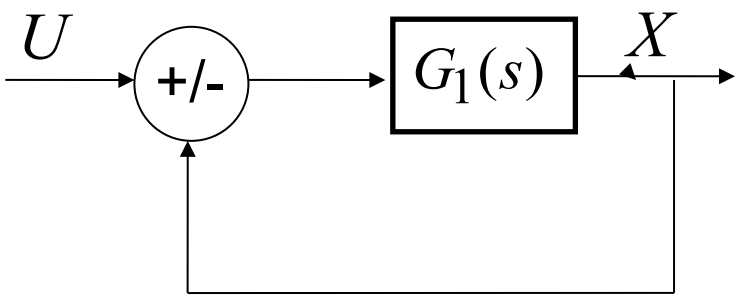
Proportional Control of Ventilation Rate

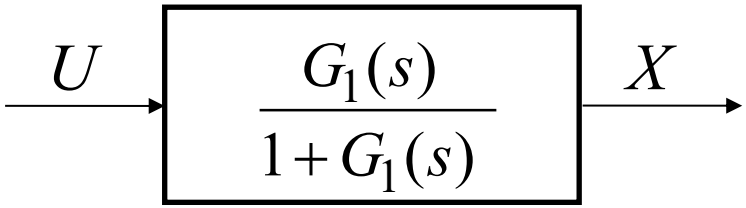


$$\frac{\left(\dfrac{1.5k_p}{10s+1}\right)}{1+\left(\dfrac{1.5k_p}{10s+1}\right)}$$

"Closed–Loop Transfer Function"

Unity Negative Feedback (Recap)





Closed–Loop Transfer Function

$$X = \frac{\left(\dfrac{1.5k_p}{10s+1}\right)}{1+\left(\dfrac{1.5k_p}{10s+1}\right)} D = \frac{1.5k_p}{10s+1+1.5k_p} D$$

What does all this mean ?

$$\tau \frac{dx}{dt} + x = Ku$$

Standard notation for 1st order system

$$\frac{K}{\tau s + 1}$$

Transfer function

$$\tau s + 1 = 0$$

"Characteristic Equation"

Characteristic Equation

$$\tau s + 1 = 0$$

"Characteristic Equation of the generalised 1st order system"

$$10s + 1 = 0$$

"Ventilation system without control"

$$10s + 1 + 1.5k_p = 0$$

"Closed–loop control system"

Hint: group $(1 + 1.5k_p)$ as a single term, and then divide through by $(1 + 1.5k_p)$

$$\left(\frac{10}{1 + 1.5k_p}\right)s + 1 = 0$$

Re-arranged to determine the new time constant – we have **changed** the behaviour of this system!

Feedback changes the dynamics of the system!

Once reduced back to "open-loop" system the same analysis can be performed, i.e. determine the stability of the system.
To do this we look at the poles of the transfer function, i.e. we find the roots of the characteristic equation, the denominator of the TF set to 0. (NB: the numerator of the TF are the zeros).

Numerator (zeros)

$$\frac{1}{(s+1)(s+2)}$$

=> Roots at -1, -2

Denominator (poles)

<u>Determine stability of closed loop system</u>

1). Rewrite as open loop (reduced block diagram).

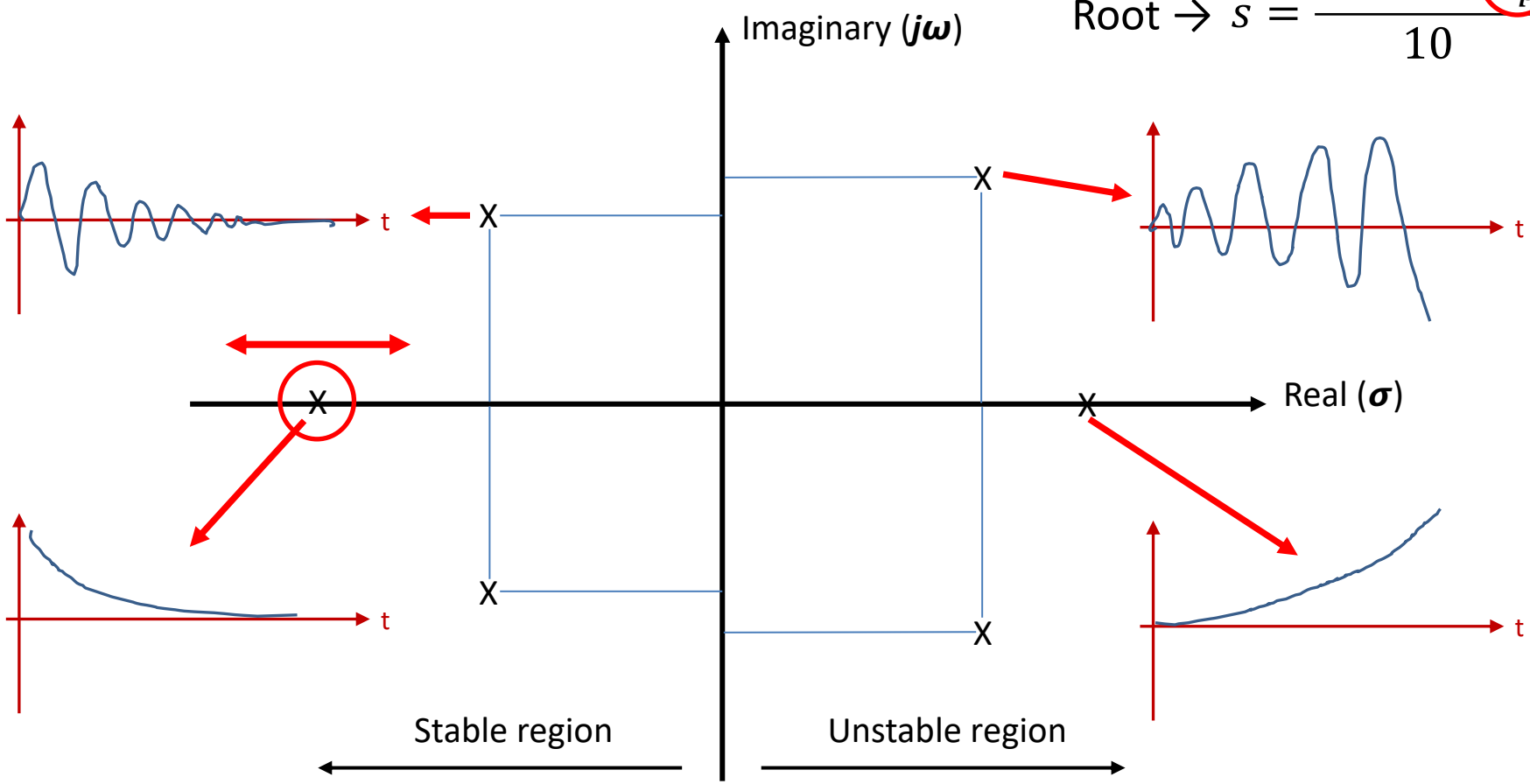
2). Solve for roots of characteristic equation.

3). Determine stability by plotting the poles in the s-plane and looking at the real components.

Relevance of the poles in the s-plane

$$10s + 1 + 1.5k_p = 0$$ ← Characteristic equation of ventilation system with closed–loop control

Root → $s = \dfrac{-1 - 1.5k_p}{10}$



Steady State Gain

$$X = \frac{1.5k_p}{10s + 1 + 1.5k_p} D$$

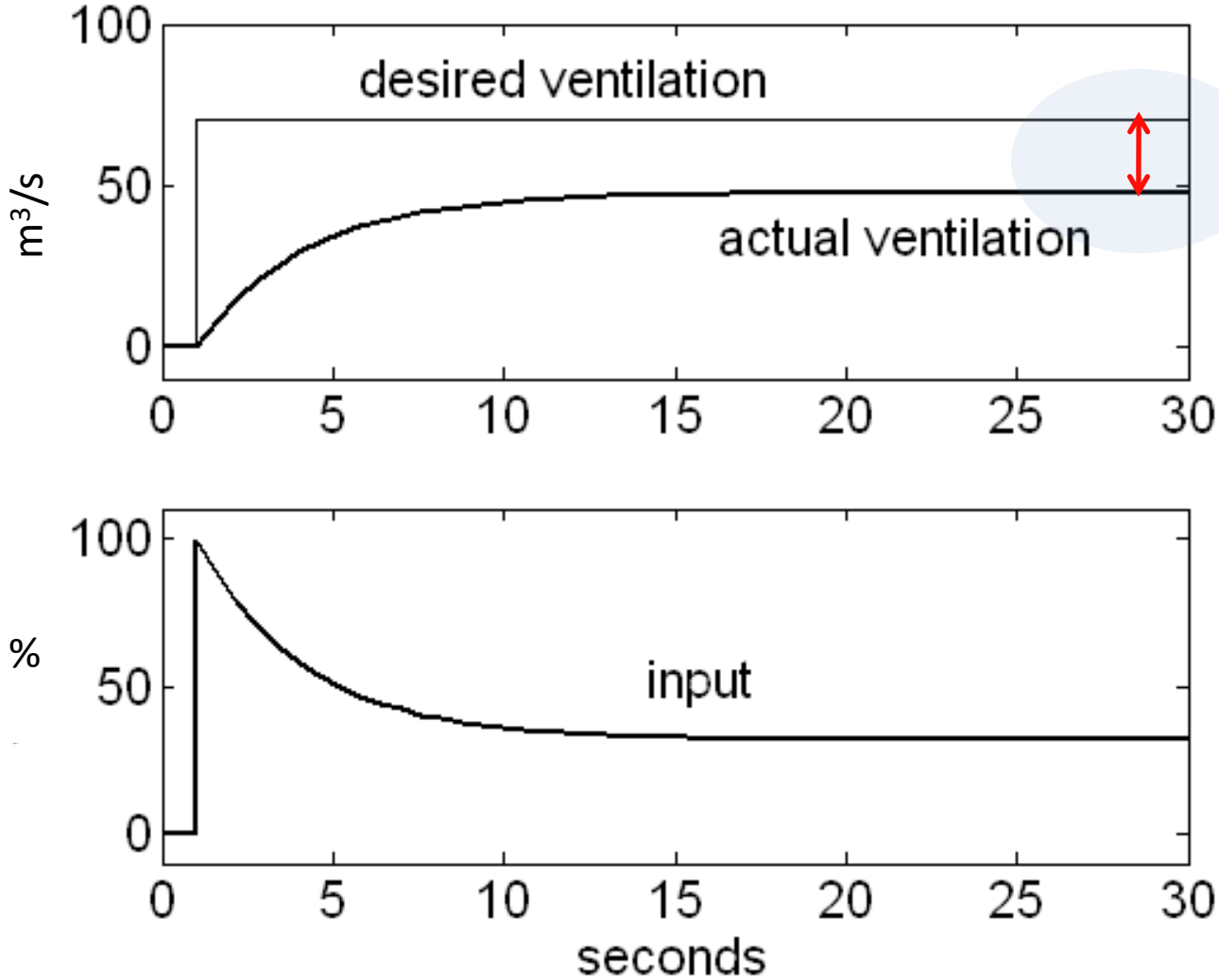
$$x(t \to \infty) = \frac{1.5k_p}{1 + 1.5k_p} d$$

"Steady State Gain"

"Steady State Value of the Output"

This is less than unity – at steady state, the actual ventilation rate will be less than the desired ventilation rate!

Proportional Control of Ventilation Rate



"Steady State Error"

Steady State Error

$$x(t \to \infty) = \frac{1.5k_p}{1+1.5k_p} d$$

$$d - x(t \to \infty) = d - \frac{1.5k_p}{1+1.5k_p} d$$

What is the steady state error for the case that $d = 1$?

$$1 - \frac{1.5k_p}{1+1.5k_p}$$

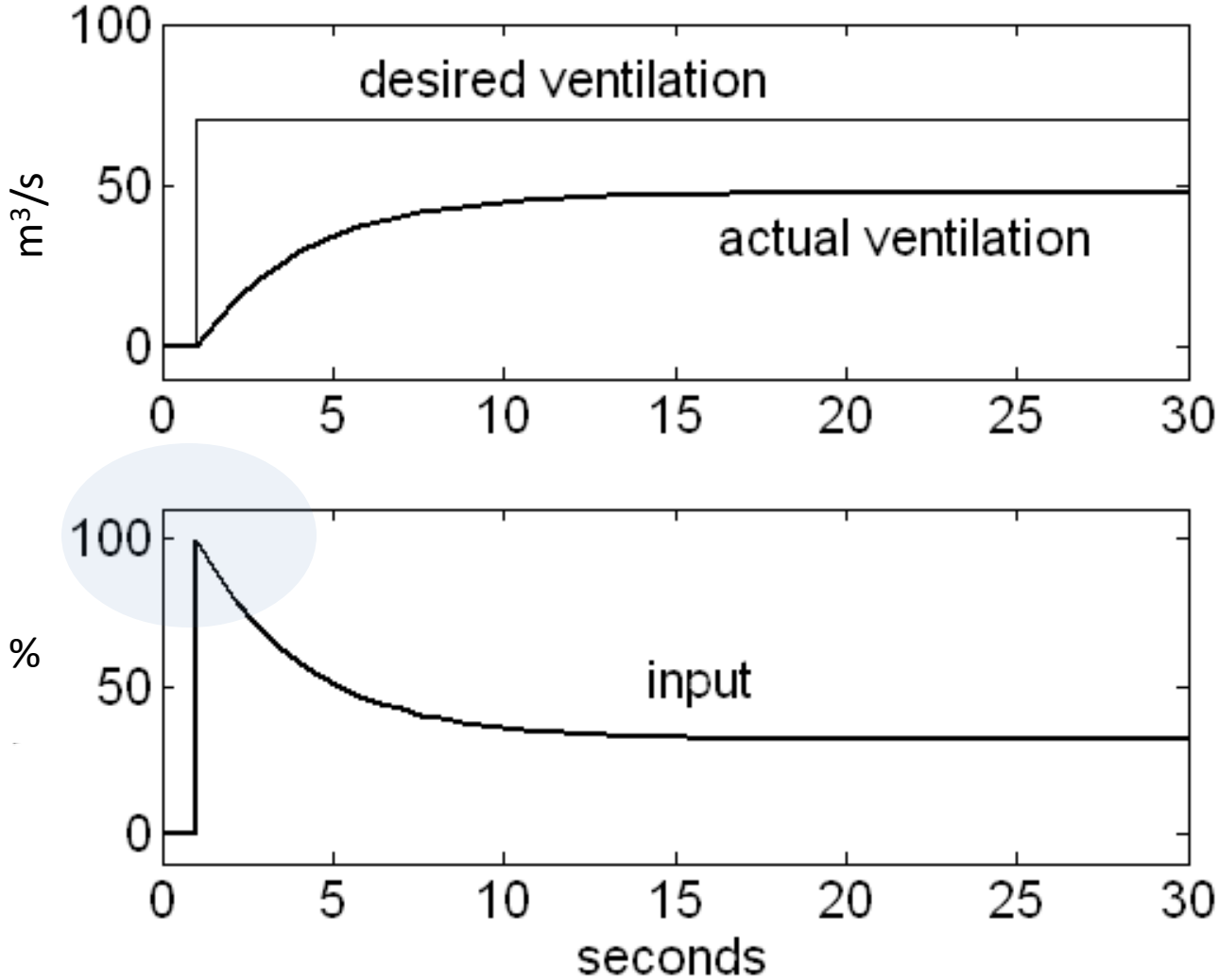
Steady State Error

$$1-\frac{1.5k_p}{1+1.5k_p}=\frac{1+1.5k_p}{1+1.5k_p}-\frac{1.5k_p}{1+1.5k_p}$$

$$=\frac{1+1.5k_p-1.5k_p}{1+1.5k_p}=\frac{1}{1+1.5k_p}$$

The larger the control gain, the smaller the steady state error …

Proportional Control of Ventilation Rate



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